Long-Term Growth Effects of Ideological Turnover in Parliamentary Democracies

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Abstract

Economic growth depends on not only on structural fundamentals, but also on whether governments change policy in stable and predictable ways. This paper asks whether larger shifts in governing ideology reduce annual GDP per capita growth in parliamentary democracies. Using Penn World Table data matched to the Comparative Political Data Set for 21 parliamentary democracies from 1962 to 2023, we estimate two-way fixed-effects growth regressions with standard controls for lagged income, savings, human capital, and population growth. In the standard model, a one-unit increase in lagged ideology-shift magnitude is associated with about a 0.861 percentage-point reduction in annual GDP per capita growth, and threshold estimates show that the penalty becomes larger as the ideological shift moves from small to medium to large. This negative relationship survives winsorisation, the exclusion of 2008 and 2009, and an event-style large-shift specification. Additionally, Granger-causality results show that past ideology shifts predict growth more strongly than past growth predicts ideology shifts. An extension suggests that large shifts in either direction are costly, with possible larger penalties for large rightward shifts. Therefore, the results support a policy-discontinuity interpretation. Specifically, abrupt ideological turnover appears to weaken growth by increasing uncertainty and adjustment costs rather than because one ideology is always superior.

I. Introduction

In the four years after France's 1981 election brought the left to power for the first time in a generation, investment collapsed and the government reversed course twice. It was the first time a Socialist has held the office since the establishment of the Fifth Republic. Additionally, there was an immediate financial panic, causing a sell-off in the stock market and a sharp drop in the franc's value which triggered a flight of capital. Thus, investors panicked and forced the government to impose capital controls and raise interest rates to protect their currency. This shows how abruptly shifting policy environments can destabilize an economy and is a pattern that gives rise to the questions about what drives long-run growth and why economies recover while others do not.

Economic growth is one of the most important subjects in economics because even a small difference in annual growth rate compound into large differences in living standards over time. In a long-run growth-accounting framework, persistent changes in productivity, savings, human capital, and demography shape the path of income per capita. However, countries that look broadly similar on these fundamentals still experience very different growth outcomes from year to year. One plausible reason is that growth also depends on whether governments change policy in stable, credible, and predictable ways. If large shifts in governing ideology produce abrupt changes in taxation, public spending priorities, labour-market rules, and regulation, then those shifts may reduce growth even when the underlying economy is otherwise similar.

This paper addresses the question of whether larger shifts in governing ideology reduce annual GDP per capita growth in parliamentary democracies. The focus is not on whether left-wing or right-wing governments are inherently better for growth. Instead, the paper studies whether the size of ideological movement itself matters. When coalition arrangements shift sharply, firms, households, and investors may face a more uncertain policy environment, delay long-lived commitments, or incur adjustments costs as rules and priorities change.

The economic logic behind the hypothesis is straightforward. Governments affect the allocation of public resources, the design of tax systems, the stringency of regulation, the structure of redistribution, and the incentives facing labour and capital. Large ideology shifts can therefore produce two related outcomes. First, they can increase policy uncertainty by making future taxes, spending, and regulation harder to predict. Second, they can impose adjustment costs when new governments reverse or redesign policies that firms and households had already planned around.

Both point toward lower capital accumulation, delayed hiring, and weaker output growth in the period following abrupt ideological turnover. In this paper, the testable implication is that larger lagged ideology shifts should be associated with lower current GDP per capita growth, with the strongest effects concentrated in large shifts rather than routine cabinet drift.

This question sits at the intersection of several literatures, but a gap remains. The growth literature explains why lagged income, savings, human capital, and population growth belong in the regression framework. The instability literature argues that unstable political environments can reduce investment and growth. The uncertainty literature shows that firms delay investment and employment when policy becomes harder to predict. A related ideology literature studies how left and right governments differ in budget composition and the broader supply side of the economy. What is less well understood is whether the size and direction of cabinet ideology shifts predict growth when they are embedded in a standard panel growth model with country and year fixed effects.

This paper contributes to the literature by connecting cabinet ideology shifts to long-run growth in a way that keeps the political mechanism transparent. Rather than treating ideology as a fixed government characteristic, the paper uses changes in cabinet ideology as a proxy for policy discontinuity, which is closer to the uncertainty and adjustment-cost channel motivating the paper. It also distinguishes between the size and the direction of ideological turnover, allowing the main result to focus on whether large shifts are more costly than routine cabinet drift before turning to whether leftward and rightward shifts differ. Finally, the paper strengthens the interpretation of the results with a Granger causality test that compares whether ideology shifts precede weaker growth more clearly than weak growth precedes ideology shifts.

The results support the main hypothesis. In the standard model, a one-unit increase in lagged ideology-shift magnitude is associated with roughly 0.861 percentage points lower annual GDP per capita growth. The threshold results indicate that the penalty grows larger as shifts become larger, and the robustness checks show that the negative association survives winsorisation, the exclusion of crisis years, and an event-style large-shift specification. The directional extension suggests that large shifts in either direction are costly, though the estimated penalty is somewhat larger for large rightward shifts.

The rest of the paper proceeds as follows. Section II reviews the growth, instability, uncertainty, and ideology literatures that motivate the project. Section III describes the data sources,

variable construction, expected signs, and limitations of the dataset. Section IV presents summary statistics and the main descriptive figures. Section V outlines the empirical model and explains why the standard model uses two-way fixed effects with panel-corrected standard errors and AR(1) correction. Section VI discusses the main regression results, threshold results, robustness checks, Granger-causality results, and the directionality extension. Section VII concludes with the paper's implications, limitations, and suggestions for future research.

II. Literature Review

A large body of research in economics seeks to explain differences in long-run economic growth across countries. This literature has traditionally emphasized structural determinants such as capital accumulation, human capital and population growth, while more recent work highlights the role of political and institutional factors. This section reviews the empirical growth literature and the political economy of growth, providing the basis for examining how changes in governing ideology may affect economic performance.

Empirical growth research builds on the standard framework in which macroeconomic fundamentals are key determinants of long-run performance. Darby, Li and Muscatelli (2004) examine the relationship between macroeconomic stability, policy credibility and economic growth using cross-country data. Their analysis shows that instability in macroeconomic policy can have persistent negative effects on growth by increasing uncertainty and discouraging investment. Similarly, Tavares (2004) studies the political determinants of economic growth, emphasizing how institutional and policy environments shape economic outcomes across countries. Using panel data, the paper finds that political conditions, including policy stability and institutional quality, play a significant role in determining growth performance. Together, these contributions reinforce the empirical importance of macroeconomic fundamentals while highlighting the role of policy environments in shaping long-run growth outcomes. This provides a foundation for including both standard economic controls and politically driven variables in growth regressions.

While early growth models focused primarily on structural determinants, subsequent work explicitly incorporates political factors into the analysis. Alesina and Perotti (1994) provide a comprehensive survey of the political economy of growth, arguing that political instability and institutional quality play a central role in shaping economic performance. Drawing on cross-country evidence, they show that unstable political environments can distort economic policy,

reduce investment and weaken long-run growth prospects. Their analysis emphasizes that economic outcomes cannot be fully understood without accounting for the political context in which policy decisions are made. This literature therefore expands the standard growth framework by identifying channels through which political conditions influence economic activity, particularly through their effects on policy credibility and investment behaviour.

A more direct empirical link between political instability and growth is established by Alesina et al. (1996), who examine how political instability affects economic performance across countries. Using cross-country panel data, they find that higher levels of political instability are associated with lower investment and reduced economic growth. Their results suggest that uncertainty about future policy direction discourages investment and disrupts economic activity, providing a clear mechanism through which political factors can influence growth outcomes. This finding is particularly relevant for understanding the effects of abrupt political changes, as large shifts in governing ideology may generate similar forms of uncertainty and adjustment costs.

Together, this literature establishes two key insights. First, long-run growth is shaped not only by macroeconomic fundamentals but also by the stability and credibility of economic policy. Second, political instability can adversely affect growth through its impact on investment and expectations. However, while existing research emphasizes political instability broadly defined, it provides less direct evidence on how specific forms of political change, such as shifts in governing ideology, affect economic performance. This gap motivates the analysis of ideology shifts as a measure of policy discontinuity, which is further developed in the subsequent literature on economic policy uncertainty and partisan policy effects.

The uncertainty effect is a direct bridge between political change and overall investment behaviour. For example, Baker Bloom and Davis (2016) construct a news-based index of economic policy uncertainty for different advanced economics and show that uncertainty is associated with declines in investment, output, and employment. Specifically, their identification strategy combines newspaper coverage of policy-related uncertainty, disagreement among forecasters about government spending, and variation in expiring tax code provisions. Then, they built a composite measure that is independent of some of the standard macroeconomic conditions. The key finding is that uncertainty spikes around major events, elections, and policy reversals. Additionally, those spikes predict contractions in investment and hiring in the following quarters. To connect this to our paper, the ideology-shift variable used here does not directly observe uncertainty, but it proxies

for the political conditions under which uncertainty may arise. The threshold results in this paper show that are penalties are significant when there are large shifts. This supports their findings that uncertainty effects are nonlinear and concentrated around major political disruptions rather than ordinary policy drift.

Furthermore, Boix (1997) studies the provision of physical and human capital across advanced democracies from 1960 to 1990 and finds that left and right governments differ in how they allocate public investment. For example, more left-wing governments tend to prioritize human capital formation through education and training expenditure, while more right-wing governments tend to favour physical infrastructure and private investment incentives. Boix interprets this as a partisan difference. In other words, left-wing parties see human capital as the binding constraint, while right-wing parties emphasize the returns to physical capital and deregulation. For our paper, this is highlighted in two ways. First, it confirms that ideology is not label but a predictor on how government allocate resources, which justifies treating ideology shifts as a proxy for political discontinuity. Second, if a firm operating under one regime faces a fundamentally different cost structure when the other takes over, then the adjustment runs in different directions depending on which way the cabinet moves.

Potrafke (2011) extends that logic by looking at whether government ideology influences budget composition in OECD countries, rather than just the stated priorities. Using a panel covering several decades, he finds that left governments are associated with higher social expenditure shares and somewhat overall spending levels and right governments are associated with lower transfer and greater emphasis on fiscal adjustment. Additionally, he shows these differences are meaningful even after controlling for electoral cycles, fiscal rules, and the business cycle. This research strengthens our policy-discontinuity interpretation. In other words, if ideology predicts budget composition, then a large ideology shift implies that not only leadership rhetoric, but fiscal priorities will change. Therefore, firms and households that have planned around one regime face adjustment costs when that regime changes abruptly.

Haini and Loon (2021) show that the relationship between government spending and growth is conditional on ideology. Using a panel of developing and developed economies, they find that the effects of public expenditure vary depending on the orientation of the government overseeing that spending. Therefore, their results suggest that ideology does not determine the level of spending but shapes how that spending translates into output. For our paper, Haini and Loon's

reinforce the argument that ideological turnover matters for growth for productive efficiency of the fiscal environment that follows a cabinet change.

III. Data Description

In this paper, we have orchestrated a country-year panel dataset to examine whether shifts in government ideology are associated with annual GDP per capita growth in parliamentary democracies. It combines macroeconomic variables from the Penn World Table 11.0 with political and institutional variables from the Comparative Political Data Set (CPDS). After cleaning and merging the datasets, our analysis sample includes 21 parliamentary democracies observed between 1962 and 2023, with 1,078 country-year observations in the main model.

A. GDP per Capita Growth and Growth Controls

The Penn World Table 11.0 (Feenstra, Inklaar, and Timmer 2025), whose construction is described in Feenstra, Inklaar, and Timmer (2015), provides GDP per capita, savings, human capital, and population. For ease of exposition, the paper refers to annual GDP per capita growth as GROWTH, lagged ideology-shift magnitude as SHIFT, lagged log GDP per capita as LGDPPC, log savings share as LSAVE, log human capital as LHC, and population growth as POPG.

GROWTH is the year-to-year log difference in GDP per capita expressed in percentage terms. LGDPPC is expected to be negative because richer countries should grow more slowly conditional on the other determinants of growth. LSAVE proxies' physical capital accumulation and is expected to be positive, while LHC captures education and skills and is also expected to be positive. POPG is the annual growth in a country's total population. It is constructed as the year-to-year log difference in population and expressed in percentage terms. POPG is expected to be negative because faster population growth can diminish capital per worker unless counteracted by stronger investment within the standard growth model.

B. Government Ideology

The Comparative Political Data Set (Armingeon et al. 2025) provides cabinet ideology, parliamentary-democracy status, and proportional-representation status for the political side of the analysis.

The key explanatory variable is SHIFT. Cabinet ideology is measured as a continuous score based on the cabinet shares classified as left and right, scaled from negative one for a fully left

cabinet to positive one for a fully right cabinet. Annual ideology shift is the year-to-year change in that score, and ideology-shift magnitude is the absolute value of that change. A value of zero therefore indicates no ideological change in cabinet composition, while larger values indicate larger ideological movement regardless of direction. The paper focuses on parliamentary democracies because cabinet composition maps more directly into governing policy in parliamentary systems than in presidential systems. The expected sign of SHIFT is negative because larger ideological shifts may increase uncertainty and adjustment costs.

C. Directional and Threshold Variables

In addition to SHIFT, the paper constructs small, medium, and large shift indicators to test whether the growth effect is concentrated in large ideological realignments rather than routine cabinet drift. It also constructs small, medium, and large leftward and rightward shifts for the directionality extension. These directional bins are not the main test, but they help assess whether the cost of ideological turnover differs by direction. Finally, the paper constructs a five-year ideology-volatility measure for the descriptive figures because the one-year shift variable is heavily concentrated at zero.

D. Sample Construction and Weaknesses

The final sample is restricted to country-years that are parliamentary democracies, have non-missing growth and standard macroeconomic controls, and have enough political information to construct lagged ideology-shift measures. The main weakness of the data is that ideology shift is a proxy for policy discontinuity rather than a direct measure of policy change. More overall, a recurring limitation in our political economy literature review is the absence of a single comprehensive dataset that directly links policy contents to economic outcomes across countries and over time. This paper faces that same limitation, in that the dataset does not directly observe tax reforms, regulation, business expectations, or investment timing, and some directional bins, especially large leftward and large rightward shifts, contain relatively few observations. The combined dataset is well suited to the paper's question because it provides a long-run panel structure, internationally comparable growth controls, and a transparent measure of ideological movement in government.

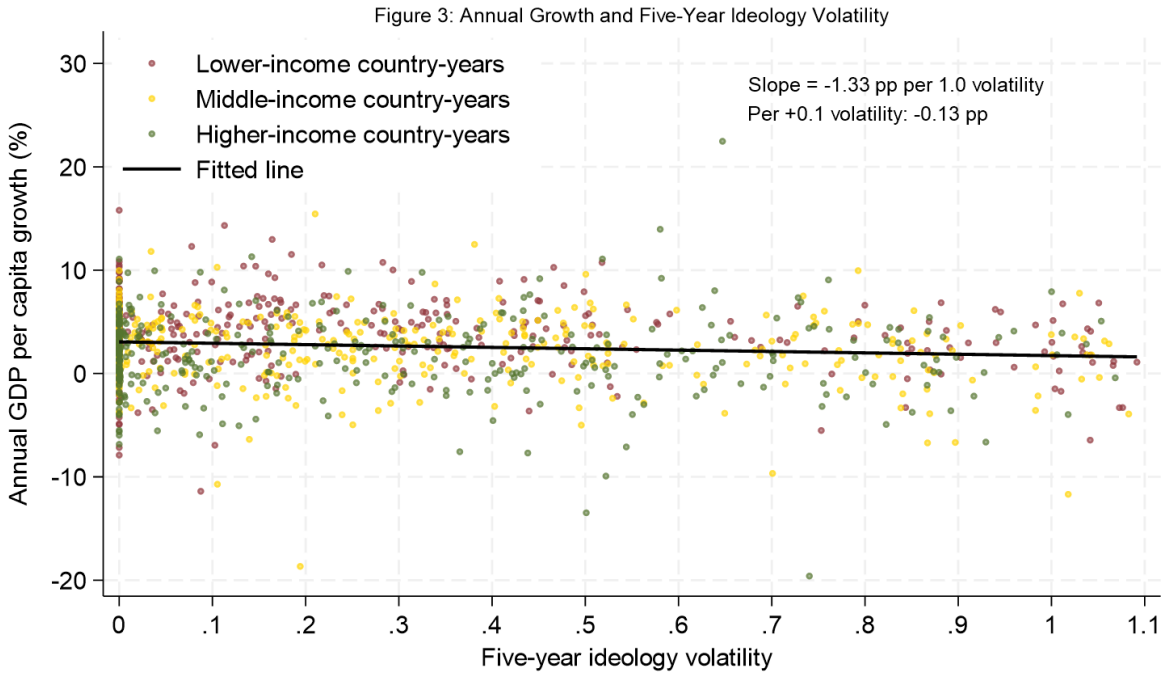
IV. Summary Statistics

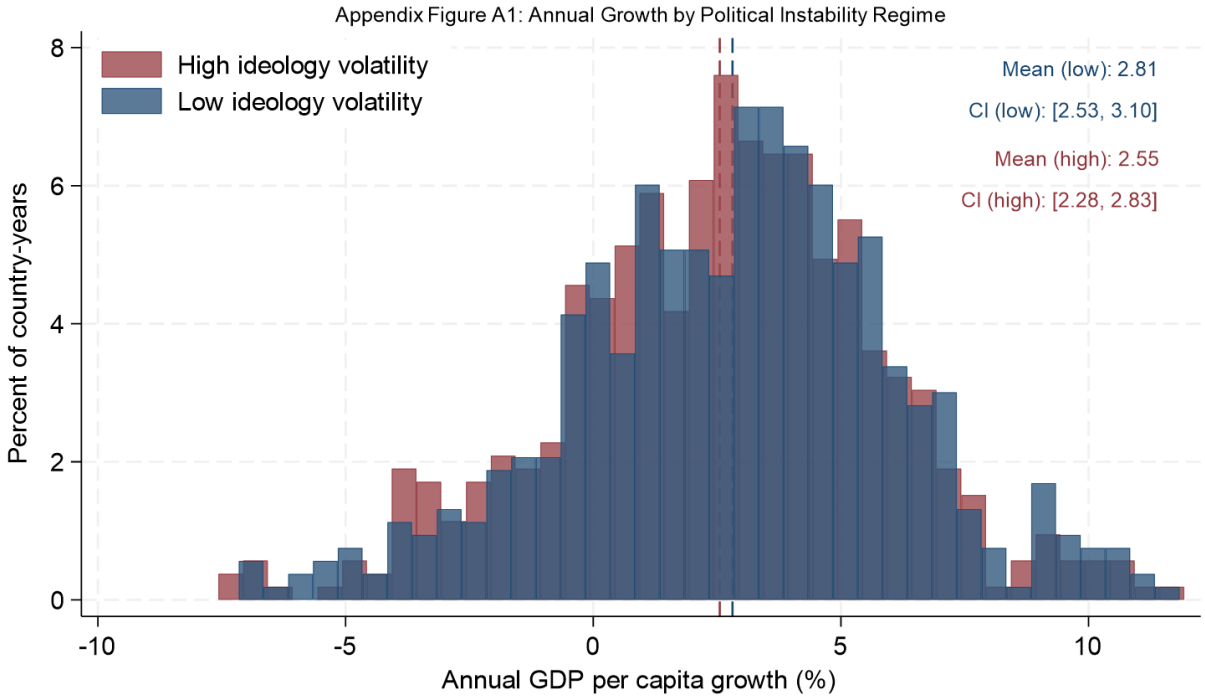
Table 1 reports summary statistics for 1998, the year with the maximum number of usable observations in the final analysis sample. The table reports the mean, standard deviation, minimum, maximum, and number of observations for the key variables using non-logged units. Figure 3 plots annual GDP per capita growth against five-year ideology volatility. The scatter does not reveal a clear pattern by income group, suggesting that the negative descriptive relationship is not plainly driven by a wealth effect in this sample. Figure A1 compares the growth distributions in low-volatility and high-volatility regimes. Collectively, these descriptive figures show that more unstable ideological environments are associated with weaker growth in the raw data, which is consistent with the paper's hypothesis but does not replace the regression analysis.

Table 1: Summary Statistics for Exploratory and Outcome Variables

Variable	Mean	SD	Min	Max	N
Annual GDP per capita growth (%)	3.312	2.844	-2.135	9.276	21
GDP per capita (thousand 2017 USD)	32.977	13.340	10.465	70.168	21
Population (millions)	23.985	32.517	0.394	126.549	21
Savings share of GDP (%)	24.178	3.764	15.768	33.115	21
Human capital index	3.123	0.303	2.602	3.539	21
Lagged ideology-shift magnitude	0.214	0.350	0.000	1.337	21

Notes: Representative year: 1998. This table reports the mean, standard deviation, minimum, maximum, and number of observations for the outcome and explanatory variables used in the paper. The table uses the final parliamentary-democracy analysis sample and non-logged variables only.





V. Empirical Strategy

The baseline empirical model accompanies the standard long-run growth-accounting framework while adding the paper's political variable of interest:

$$\Delta Y_{i,t} = \beta_0 + \beta_1 SHIFT_{i,t-1} + \beta_2 \ln Y_{i,t-1} + \beta_3 \ln S_{i,t} + \beta_4 \ln HC_{it} + \beta_5 \Delta lnP_{i,t} + \mu_i + \lambda_t + \epsilon_{i,t}$$

In this equation, $\Delta Y_{i,t}$ is annual GDP per capita growth, $SHIFT_{i,t-1}$ is lagged ideology-shift magnitude, $lnY_{i,t-1}$ is lagged log GDP per capita, $lnS_{i,t}$ is log savings share of GDP, $lnHC_{i,t}$ is log human capital, and $\Delta lnP_{i,t}$ is population growth in log-difference form. The term μ_i captures country fixed effects and λ_t captures year fixed effects. Country fixed effects capture time-invariant national attributes such as geography, long-run institutions, or persistent political traditions. Year fixed effects absorb global shocks shared across countries, such as recessions, commodity shocks, or worldwide financial disturbances.

The dependent variable is annual GDP per capita growth. The central question of the paper is whether ideological turnover is followed by weaker economic performance, hence this measure captures the outcome of interest directly. Using annual growth allows the analysis to test whether large shifts in governing ideology are associated with lower GDP per capita growth in the following

year. The main political variable is therefore lagged by one year, since policy shifts are unlikely to affect measured growth instantaneously. This lag structure also allows for cleaner interpretation and aligns with the Granger-causality test reported later in the paper.

The preferred estimator is a Prais-Winsten-style panel regression with panel-corrected standard errors that allow for heteroskedasticity across panels and AR(1) serial correlation within panels, consistent with the post-regression diagnostics. The country fixed-effects test strongly supports including country effects, the Hausman test rejects random effects in favour of fixed effects, and the year fixed-effects test supports including common year shocks. The modified Wald test detects heteroskedasticity, the Wooldridge-style serial-correlation test detects within-country serial correlation, and the quadratic ideology-shift test does not support keeping a nonlinear squared term. Therefore, the preferred estimator corrects the two problems that are present in the data without unnecessarily complicating the model.

More specifically, the diagnostics indicate the following: the country fixed-effects test produces $F(13,20) = 3130.07$ with $p = 0.0000$; the Hausman test gives $chi2(13) = 60.12$ with $p = 0.0000$; the year fixed-effects test gives $F(61,991) = 7.88$ with $p = 0.0000$; the heteroskedasticity test gives $chi2(21) = 609.31$ with $p = 0.0000$; the serial-correlation test gives $F(1,20) = 225.248$ with $p = 0.0000$; and the quadratic ideology-shift term has $p = 0.8165$. These results mutually imply that the correct baseline is a two-way fixed-effects model with corrected standard errors, not pooled OLS, not random effects, and not a nonlinear ideology specification.

This paper estimates a sequence of complementary models to determine whether ideological turnover is correlated with weaker subsequent growth, whether that relationship is concentrated in large shifts, whether it survives alternative specifications, and whether it reflects the size of the shift more than its direction. Table 2 begins with the baseline two-way fixed-effects regressions and adds the standard growth controls step by step to show that the negative association between ideology shifts and growth is not simply due to omitted convergence, savings, human capital, or population effects. Table 3 then replaces the continuous shift measure with small, medium, and large lagged shift indicators to test the paper's core mechanism, namely whether large ideological realignments are more costly than routine cabinet drift. Table 4 examines whether the result is robust to outliers, crisis years, and an alternative event-style coding of major shifts. Table 5 addresses whether the observed relationship is more consistent with ideology shifts preceding weaker growth than with weak growth preceding ideology shifts through a Granger causality test.

Finally, Table 6 and Table 7 extend the analysis by asking whether the relationship differs by direction and by electoral system, which helps clarify whether the main result reflects a general cost of policy discontinuity or a more specific institutional pattern.

VI. Results and Discussion

A. Main Results

Table 2 reports the main stepwise two-way fixed-effects regressions and provides the main result for the paper. In the standard model, the coefficient on lagged ideology-shift magnitude is -0.861 and statistically significant at the 1 percent level. This means that, holding constant lagged GDP per capita, savings, human capital, population growth, country fixed effects, and year fixed effects, a larger ideology shift in one year is associated with lower GDP per capita growth in the next year. The coefficient remains negative and significant in every stepwise column, beginning with the limited model and continuing as the standard growth controls are added. This stability indicates that the estimated relationship is not fragile to the inclusion of additional controls. Instead, ideological turnover remains negatively associated with growth even after accounting for the main economic determinants typically included in long-run growth regressions.

Table 2: Growth on Ideology Shifts with TWFE

TWFE	Annual growth rate				
	(1)	(2)	(3)	(4)	(5)
Lagged ideology-shift magnitude	-0.959***	-0.959***	-0.860***	-0.869***	-0.861***
	(0.278)	(0.270)	(0.268)	(0.268)	(0.268)
Lagged log GDP per capita		-5.073***	-6.360***	-7.100***	-7.547***
		(0.931)	(0.963)	(1.047)	(1.077)
Log savings share of GDP			4.553***	4.546***	4.323***
			(0.894)	(0.896)	(0.930)
Log human capital				5.499	5.829
				(3.750)	(3.776)
Population growth (%)					0.382
					(0.337)
Year FE	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓
Observations	1078	1078	1078	1078	1078
R-Squared	0.37	0.40	0.42	0.42	0.42

Note: Sample restricted to parliamentary democracies, 1962-2023. Standard Errors are panel-corrected and allow for heteroskedasticity and AR(1) serial correlation.
*** p<0.01, ** p<0.05, * p<0.1

The control variables also behave largely as expected. Lagged log GDP per capita is negative and significant, which is consistent with conditional convergence. Log savings share is positive and significant, which fits the role of capital accumulation in the growth model. Human capital is positive but not precisely estimated in the standard model, and population growth is positive but not statistically significant. The R-squared also rises from 0.37 to 0.42 as the controls are added. This suggests that the standard growth variables do meaningful work in the model, while also leaving room for ideology shifts to explain an economically meaningful portion of within-country variation in annual growth.

Table 3 and Figure 1 sharpen the mechanism by showing that the effect is concentrated in larger shifts. Figure 1 presents mean annual GDP per capita growth by shift-size category. Growth is highest in no-shift years and lowest after large ideology shifts, with the small and medium categories lying between those two extremes. Table 3 spells out that pattern. Relative to the omitted no-shift category, small ideology shifts are associated with about 0.533 percentage points lower growth, medium ideology shifts with about 0.847 percentage points lower growth, and large ideology shifts with about 1.413 percentage points lower growth. The key point is that the penalty grows with the size of the ideological realignment, which is exactly what a policy-discontinuity interpretation would predict.

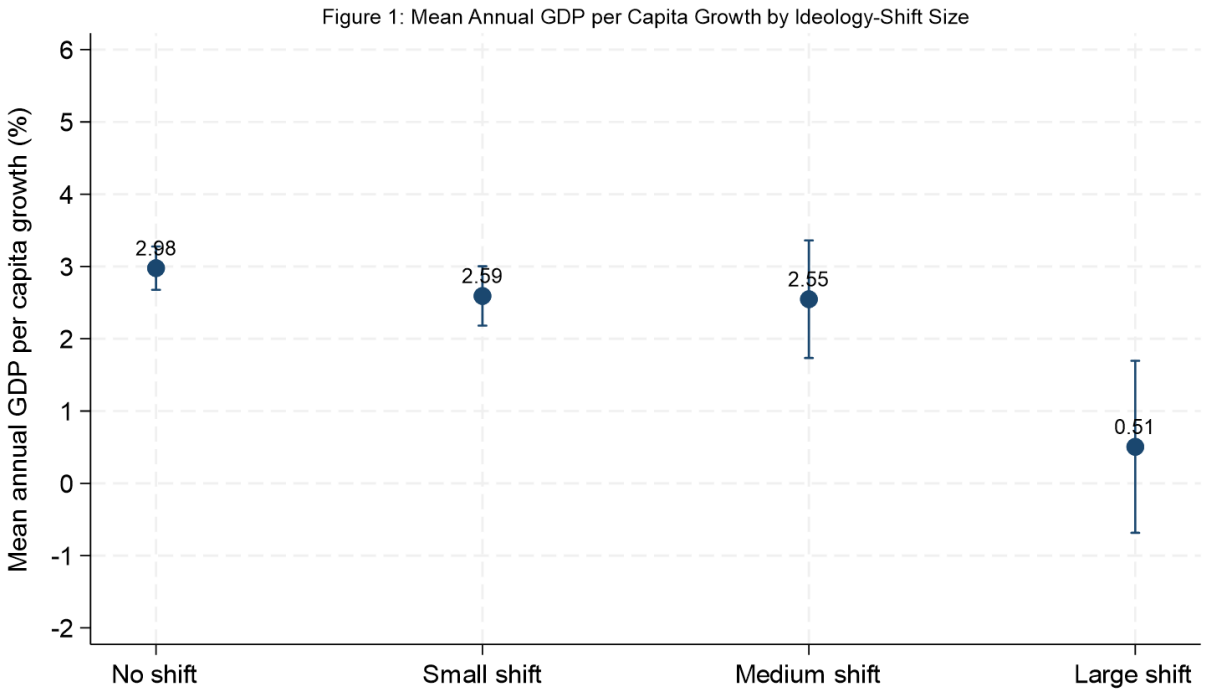


Table 3: Threshold Regressions for Small, Medium, and Large Ideology Shifts

	Annual growth rate
Lagged small ideology shift	-0.533**
	(0.238)
Lagged medium ideology shift	-0.847**
	(0.368)
Lagged large ideology shift	-1.413***
	(0.456)
Lagged log GDP per capita	-7.458***
	(1.073)
Log savings share of GDP	4.340***
	(0.926)
Log human capital	5.662
	(3.760)
Population growth (%)	0.353
	(0.337)
Year FE	✓
Country FE	✓
Observations	1078
R-Squared	0.42

Note: This mechanism table tests whether the growth effect comes from large ideology shifts rather than minor cabinet drift. Coefficients are relative to the omitted category of no ideological shift (baseline). Sample restricted to parliamentary democracies, 1962-2023. Standard Errors are panel-corrected and allow for heteroskedasticity and AR(1) serial correlation.
*** p<0.01, ** p<0.05, * p<0.1

B. Robustness and Granger-Causality Results

Figure 2 provides additional descriptive support by tracing growth around large ideology-shift event. Growth is weakest around the years following major ideological turnover and then recovers later. This figure should not be interpreted as a clean event-study design because it does not solve the full identification problem on its own. Its value is descriptive as it shows that the timing of growth slowdowns lines up with the idea that abrupt ideological realignment produces disruption and adjustment costs rather than a permanent collapse in underlying fundamentals. That visual evidence fits the threshold table especially well. The slowdown appears around the years when ideology movement is largest, which reinforces the paper's claim that the main issue is discontinuity, not ideological movement per se.

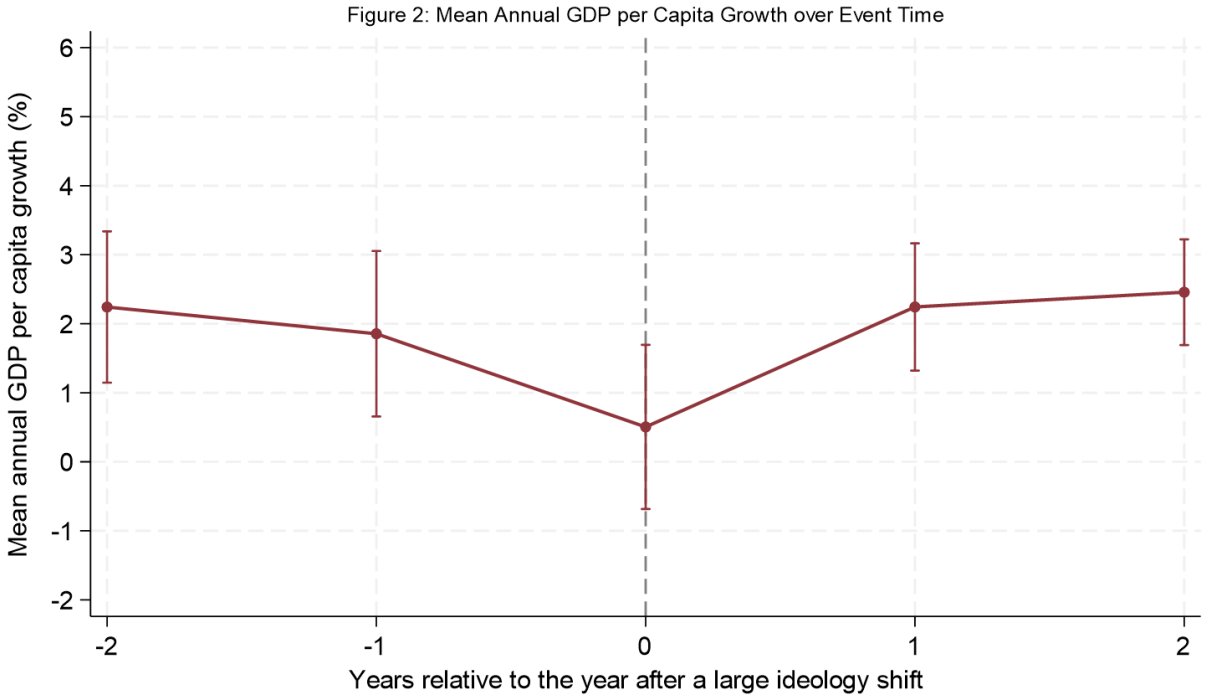


Table 4 examines whether the main relationship survives alternative specifications. Column 1 reports the standard corrected fixed-effects model. Column 2 uses a winsorised specification in which key regression variables are capped at the 1st and 99th percentiles to reduce the influence of extreme outliers without dropping observations. Column 3 removes the global financial crisis years 2008 and 2009. Column 4 replaces the continuous shift measure with a lagged large-shift event indicator. Across all four columns, the coefficient remains negative. These checks show that the core finding is not driven by a few extreme observations, by the crisis years alone, or by one coding of the ideology variable.

Table 4: Fixed-Effects Robustness Checks

	Annual Growth Rate			
	Standard Model	Standard Model w/ winsorized variables	Standard Model w/o crisis years	Standard Model w/ lagged large-shift specification
Lagged ideology-shift magnitude	-0.861*** (0.268)		-0.935*** (0.275)	
Lagged large ideology-shift event				-1.131** (0.444)
Lagged log GDP per capita	-7.547*** (1.077)		-7.394*** (1.077)	-7.490*** (1.073)
Log savings share of GDP	4.323*** (0.930)		4.364*** (0.935)	4.384*** (0.926)
Log human capital	5.829 (3.776)		5.677 (3.829)	5.560 (3.766)
Population growth (%)	0.382 (0.337)		0.328 (0.341)	0.397 (0.336)
Winsorized lagged ideology-shift magnitude		-0.685*** (0.238)		
Winsorized lagged log GDP per capita		-6.902*** (0.982)		
Winsorized log savings share of GDP		4.240*** (0.828)		
Winsorized log human capital		5.448 (3.379)		
Winsorized population growth (%)		0.139 (0.308)		
Year FE	✓	✓	✓	✓
Country FE	✓	✓	✓	✓
Observations	1078	1078	1038	1078
R-Squared	0.42	0.46	0.40	0.42

Note: Column 1 reports the standard corrected FE model. Column 2 reports the winsorized specification, in which the key regression variables are capped at the 1st and 99th percentiles to reduce the influence of extreme outliers without dropping observations. Column 3 excludes the global financial crisis years 2008 and 2009. Column 4 uses a lagged large ideology-shift event indicator. A large ideology-shift event equals 1 when the previous year's ideology-shift magnitude is at least 1. Sample restricted to parliamentary democracies, 1962-2023. Standard Errors are panel-corrected and allow for heteroskedasticity and AR(1) serial correlation.

*** p<0.01, ** p<0.05, * p<0.1

Table 5 then addresses reverse-causality concerns through a Granger causality test. In the first column, lagged ideology shifts are used to explain current growth after controlling for past growth and the standard growth controls, while in the second column, the direction of the test is reversed so that lagged growth is used to explain current ideology shifts after controlling for past ideology shifts. In the growth equation, lagged ideology-shift magnitude is negative and statistically significant at the 1 percent level, while the two-year lag is not significant. In the ideology-shift equation, the lagged growth terms are not statistically significant. The joint tests tell the same story: lagged ideology shifts help predict growth, whereas lagged growth does not. This is not proof of causality, but it does weaken a simple reverse-causality interpretation in which poor growth merely causes political turnover.

Table 5: Granger Reverse Causality Test

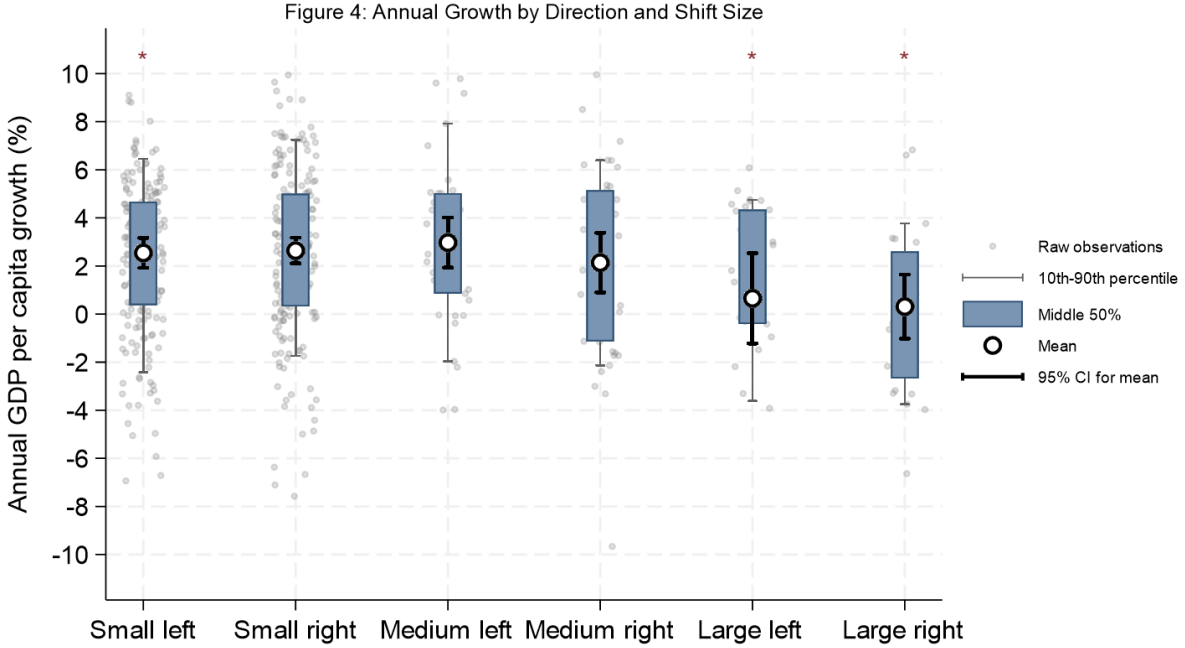
Variables	Dependent variable	
	(1) Growth	(2) Ideology shift
Lagged annual growth rate	0.182***	-0.004
	(0.048)	(0.003)
2 years lagged annual growth rate	-0.039	0.001
	(0.043)	(0.003)
Lagged ideology shift magnitude	-0.993***	0.083**
	(0.354)	(0.034)
2 years lagged ideology shift magnitude	0.188	-0.168***
	(0.185)	(0.019)
Lagged log GDP per capita	-6.113***	-0.026
	(0.740)	(0.111)
Log savings share of GDP	2.777	-0.130
	(1.622)	(0.068)
Log human capital	2.330	0.202
	(3.300)	(0.408)
Population growth (%)	0.291	-0.027
	(0.424)	(0.034)
Year FE	✓	✓
Country FE	✓	✓
Observations	1057	1057
R-Squared	0.43	0.09

Note: Column 1 asks whether past ideology shifts predict current growth. Column 2 asks whether past growth predicts current ideology movement. Sample restricted to parliamentary democracies, 1962-2023. Standard Errors are clustered at the country level.
*** p<0.01, ** p<0.05, * p<0.1

C. Comparison of Results

The directionality extension in Table 6 and Figure 4 reveals a more detailed pattern, while still supporting the broader magnitude-first conclusion. The main message remains that both leftward and rightward ideological movement can be costly. Small leftward shifts are negative and statistically significant (-0.582**), small rightward and medium rightward shifts are negative and marginally significant (-0.478*, -0.918*), and the largest penalties appear in the large leftward and large rightward categories (-1.298**, -1.559**). The large rightward coefficient is somewhat more negative than the large leftward coefficient, which suggests that large rightward realignments may be associated with slightly stronger growth penalties. However, as the confidence intervals still overlap and the directional bins contain relatively small cell sizes, this result is better interpreted as an extension rather than as a new headline finding. The safe interpretation is that large ideological movement in either direction is costly, with suggestive evidence that large rightward shifts may be more disruptive in this sample.

Figure 4: Annual Growth by Direction and Shift Size



Note: Categories are based on lagged ideology shifts.
Red * above the coresponding box plots indicates regression significance at the 5% level.

Table 6: Directional Extension: Leftward and Rightward Ideology Shifts

	Annual growth rate
	All Parliamentary Democracies
Lagged small leftward ideology shift	-0.582**
	(0.278)
Lagged small rightward ideology shift	-0.478*
	(0.281)
Lagged medium leftward ideology shift	-0.771
	(0.502)
Lagged medium rightward ideology shift	-0.918*
	(0.492)
Lagged large leftward ideology shift	-1.298**
	(0.609)
Lagged large rightward ideology shift	-1.559**
	(0.639)
Lagged log GDP per capita	-7.444***
	(1.073)
Log savings share of GDP	4.333***
	(0.928)
Log human capital	5.637
	(3.760)
Population growth (%)	0.356
	(0.338)
Year FE	✓
Country FE	✓
Observations	1078
R-Squared	0.42

Note: This directional extension splits ideology shifts into small, medium, and large leftward and rightward movements. Coefficients are relative to the omitted category of no ideological shift (baseline). Sample restricted to parliamentary democracies, 1962-2023. Standard Errors are panel-corrected and allow for heteroskedasticity and AR(1) serial correlation.
*** p<0.01, ** p<0.05, * p<0.1

Table 7 offers an additional comparison by electoral system. The main negative relationship remains strongest in parliamentary democracies with proportional representation, where the coefficient is about -0.957 and highly significant. The non-PR sample produces a coefficient in the same direction, but it is not statistically significant. This pattern should not be overstated, but it is still informative because it suggests that ideology shifts may be more clearly translated into coalition bargaining, cabinet composition, and observable policy direction in PR systems. Together, the directional and subset comparisons reinforce the wider conclusion that the main relationship is robust, even if its strength varies across institutional settings.

Table 7: Appendix Subset Comparison by Electoral System

	Annual growth rate		
	All Parliamentary Democracies	Parliamentary Democracies with proportional representation	Parliamentary Democracies without proportional representation
Lagged ideology-shift magnitude	-0.861***	-0.957***	-0.640
	(0.268)	(0.363)	(0.345)
Lagged log GDP per capita	-7.547***	-7.201***	-8.700***
	(1.077)	(1.367)	(1.459)
Log savings share of GDP	4.323***	3.851***	4.600***
	(0.930)	(1.070)	(1.597)
Log human capital	5.829	4.548	-7.237
	(3.766)	(4.764)	(6.740)
Population growth (%)	0.382	0.255	0.545
	(0.337)	(0.406)	(0.534)
Year FE	✓	✓	✓
Country FE	✓	✓	✓
Observations	1078	774	304
R-Squared	0.42	0.45	0.55

Note: Column 1 reports the full parliamentary-democracy sample. Column 2 reports parliamentary democracies with proportional representation. Column 3 reports parliamentary democracies without proportional representation. Sample restricted to parliamentary democracies, 1962-2023. Standard Errors are panel-corrected and allow for heteroskedasticity and AR(1) serial correlation.
*** p<0.01, ** p<0.05, * p<0.1

D. Broader Implications

The results show a consistent pattern between ideological shift and annual GDP per capita growth. The main fixed-effects model illustrates a negative and statistically significant relationship, which was further refine by the threshold models to show that the penalty grows as ideology shifts become larger. The robustness checks indicate that the result survives alternative specifications, and the Granger causality test weakens the possibility of a simple reverse-causality explanation. The directional extension then adds a more nuanced finding by suggesting that both large leftward and large rightward shifts are costly, with somewhat larger penalties for large rightward shifts.

These findings have broader implications for how political change should be understood in a growth context. The results suggest that growth is affected not only by the level of policy, but

also by the stability and predictability of policy transitions. If large ideological shifts create uncertainty over taxation, spending, regulation, and coalition priorities, then firms and households may delay investment, hiring, and long-term planning. With better data, direct measures of policy reform, expectations, and investment would allow this mechanism to be tested more directly. Even so, the paper's results support an interpretation in which the economic cost of political turnover lies chiefly in discontinuity itself rather than in any claim that one ideology is always superior.

VII. Conclusion

This paper shows that the economic consequences of political change depend not only on the direction of policy, but on the scale of that change. Using panel data on parliamentary democracies from 1962 to 2023, the results consistently indicate that larger shifts in governing ideology are associated with lower subsequent GDP per capita growth. In the standard model a one-unit increase in lagged ideology-shift magnitude reduces annual growth by approximately 0.861 percentage points and the threshold results demonstrate that this penalty increases sharply as ideological movement becomes larger. These findings remain robust across alternative specifications, including winsorised samples, the exclusion of crisis years and an event-style large-shift design. The Granger-causality results further suggest that past ideology shifts predict growth more strongly than past growth predicts ideology shifts, weakening a simple reverse-causality interpretation.

These results are consistent with the wider literature linking political instability and economic performance. Alesina and Perotti (1994) and Alesina et al. (1996) show that instability affects growth through its impact on investment and policy credibility and the evidence presented here aligns closely with that mechanism. Baker, Bloom and Davis (2016) demonstrate that policy uncertainty can reduce investment and hiring and the present findings suggest that abrupt ideological turnover may generate similar uncertainty even when it is not directly observed. At the same time, this paper extends the ideology literature represented by Boix (1997), Potrafke (2011) and Haini and Loon (2021) by shifting the focus away from whether left or right-wing governments perform better and toward the consequences of large ideological realignments. In that sense, the main contribution is to highlight that the disruption associated with policy discontinuity itself is an important determinant of growth outcomes.

At the same time, several limitations should be acknowledged. The ideology-shift variable is a proxy for policy discontinuity rather than a direct measure of policy change and the analysis does not observe firm-level investment, expectations or the specific fiscal and regulatory adjustments that follow political transitions. Although the Granger-causality results weaken concerns about reverse causality, the empirical strategy does not establish full causal identification. In addition, the directional extension relies on smaller sample sizes in the large-shift categories, so those estimates should be interpreted with caution. These limitations do not overturn the main findings, but they define the scope within which the results should be interpreted.

The policy implications of these findings centre on the importance of predictable and credible policy transitions. The results do not imply that political alternation should be avoided, nor that any particular ideology is inherently growth-enhancing. Rather, they suggest that the costs of political change arise when transitions are abrupt and poorly anticipated. In such environments, uncertainty over taxation, public spending and regulation can delay investment, disrupt hiring and weaken economic activity. Institutions that reduce these frictions, such as transparent fiscal frameworks, clearly communicated policy agendas and gradual implementation of major reforms, may help limit the growth costs associated with large ideological shifts. The objective is not to eliminate political change, but to reduce the economic uncertainty that accompanies it.

Future research can build on this analysis by linking ideology shifts more directly to observable policy changes, including tax reforms, expenditure composition and sector-specific regulation. Incorporating firm-level data or measures of business expectations would also allow the uncertainty mechanism to be tested more directly. In addition, further work could examine whether the effects of ideological turnover differ across policy domains or institutional settings, particularly in environments where policy commitments are more or less credible. Even with these limitations, the evidence presented here suggests that large ideological realignments are systematically associated with weaker subsequent growth and that the economic cost of political turnover lies primarily in the disruption created by policy discontinuity rather than in the inherent superiority of any particular ideology.

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